Mohammad Rohaan 22I-2327 Assignment# 02 Sec-A

PART-A Solution:

The Lagrange interpolating polynomial using 9 nodes gives a reasonably accurate approximation of the test function. The MAE indicates moderate interpolation accuracy, with slight oscillations between node points.

Lagrange Mean Absolute Error: 0.0006177047540672618.

$$\text{MAE}=\frac{1}{N}\sum_{i=1}^{N}|y_{true} - y_{pred}|$$

PART-B Solution:

The Newton divided-differences method produces the exact same interpolating polynomial as Lagrange, so the MAE is identical or extremely close. However, the Newton form is more efficient and numerically stable because it uses nested multiplication.

Newton (divided differences) MAE: 0.0006177047540672123

PART-C Solution:

Both the forward and backward Newton formulas give similar accuracy because the nodes are equally spaced, and both approximate the same degree-8 interpolating polynomial. The forward formula is slightly more accurate near the left boundary, while the backward formula is slightly more accurate near the right boundary.

Newton forward MAE: 0.0006177047540671982

Newton backward MAE: 0.0006177047540672132

PART-D Solution:

The linear spline provides a continuous but non-smooth approximation, with sharp corners at each node because the derivative changes abruptly. The natural cubic spline is much smoother

(C^2 continuous) and fits the function far better. As expected, the natural cubic spline gives a significantly lower MAE than the linear spline.

Linear spline MAE: 0.03157188544478293

Natural cubic spline MAE: 0.003316204988207245

PART-E Solution:

The degree-5 least-squares polynomial provides a smooth approximation of the data without forcing an exact match at all nodes. Its MAE is generally lower than high-degree global interpolation but slightly higher than the natural cubic spline. Because the true function is smooth but oscillatory, a degree-5 polynomial captures the general shape but cannot perfectly match the small oscillations.

Degree-5 polynomial LS MAE: 0.04009193028984285

PART-F Solution:

The fitted exponential model gives a significantly higher error than the spline and polynomial fits. This is expected because the true function contains a sinusoidal term, so a pure exponential Ae^{Bx} cannot capture the oscillations. The model only fits the general decay pattern but fails to follow the wave-like behavior of the function.

Exponential Fit A, B: 1.5620106442589403 -0.3563182638080479

Exponential Fit MAE: 0.34012661819402823

PART-G.1 Solution:

For three nodes (x_o, y_o), (x_1, y_1), (x_2, y_2), the Newton interpolating polynomial is:

$P_2(x) = c_o + c_1 (x - x_o) + c_2 (x - x_o)(x - x_1)$

Step 1 — Solve for c_o:
At $x = x_o$: $P_2(x_o) = c_o = y_o = y[x_o]$.

Step 2 — Solve for c_1:
At $x = x_1$: $P_2(x_1) = y_1 = c_o + c_1(x_1 - x_o)$.
So, $c_1 = (y_1 - y_o)/(x_1 - x_o) = y[x_o, x_1]$.

Step 3 — Solve for c_2:
At $x = x_2$: $P_2(x_2) = y_2 = c_o + c_1(x_2 - x_o) + c_2(x_2 - x_o)(x_2 - x_1)$.
Solving yields $c_2 = y[x_o, x_1, x_2]$.

Thus the quadratic Newton interpolant is:
$P_2(x) = y[x_o] + y[x_o, x_1](x - x_o) + y[x_o, x_1, x_2](x - x_o)(x - x_1)$.

PART-G.2 Solution:

For equally spaced nodes $x_i = x_o + ih$ and $u = (x - x_o)/h$, the Newton forward polynomial is:

$P_k(x) = \Sigma$ (from m=0 to k) $[\ \Delta^m y_o / m!\ ] \cdot u(u-1)...(u-m+1)$

Key property:
If $y_i = p(x_i)$ for a polynomial p(x) of degree $d \le k$, then higher forward differences vanish:
$\Delta^{d+1} p(x_i) = 0$.

Thus the Newton forward series becomes finite and exact:
$p(x) = \Sigma$ (from m=0 to d) $[\ \Delta^m p(x_o) / m!\ ] \cdot u(u-1)...(u-m+1)$.

Since $P_k(x)$ uses all non-zero terms and p(x) is also a degree ≤ k polynomial fitting the same k+1 points, by uniqueness of polynomial interpolation $P_k(x) \equiv p(x)$.

Therefore, the Newton forward polynomial of order k exactly reproduces any polynomial of degree ≤ k.

Method	MAE
Lagrange	0.0006177047540672618
Newton (Divided Differences)	0.0006177047540672123
Newton Forward	0.0006177047540671982
Newton Backward	0.0006177047540672132
Linear Spline	0.03157188544478293
Natural Cubic Spline	0.003316204988207245
Degree-5 LS Polynomial	0.04009193028984285
Exponential Fit A, B	1.5620106442589403 -0.3563182638080479
Exponential Fit MAE	0.34012661819402823